

# Douglas Ezra Morrison

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**Methodological interests:** causal inference, incomplete data, time-to-event models, interval censored data, incidence estimation, transportability, data fusion, systems modeling, statistical process control, machine learning, Bayesian methods, computational statistics, data visualization, interactive data dashboards, statistical software development.

**Application interests:** infectious diseases, neurodegenerative diseases, cancer, diagnostics, biomarkers, health services research, health care quality improvement, implementation science, mHealth.

## Education

- 2015 – 2021     **PhD, Biostatistics**, University of California, Los Angeles. Dissertation: *Statistical Challenges in Incidence Estimation using Cross-Sectional Data and Multi-Biomarker Assay Algorithms*. Advisor: Dr. Ron Brookmeyer. Conferred, September 10, 2021.
- 2009 – 2010     **MS, Statistics**, Stanford University.
- 2005 – 2009     **BS, Symbolic Systems** (neuroscience concentration), Stanford University.

## Employment

- 2022 – present     **Assistant Professor/Assistant Professor in Residence**, Division of Biostatistics, Department of Public Health Sciences, University of California, Davis, CA.
- 2021 – 2022     **Postdoctoral Scholar**, with Dr. Onyebuchi Arah, *Epidemiology*, and Dr. Moira Inkelas, *Health Policy & Management*, UCLA: COVID-19 surveillance and mitigation in the K-12 setting.
- 2020 – 2021     **Graduate Student Researcher**, with Dr. Moira Inkelas, *Health Policy & Management*, Dr. Onyebuchi Arah, *Epidemiology*, and Dr. Roch Nianogo, *Epidemiology*, UCLA: Agent-based models of school reopening strategies during the COVID-19 pandemic.
- 2016 – 2021     **Graduate Student Researcher**, with Dr. Tom Belin, *Biostatistics Department*, and Dr. Vivek Shetty, *School of Dentistry*, UCLA: Development and assessment of mobile health apps for monitoring and promoting oral health behaviors.
- 2015 – 2021     **Graduate Student Researcher**, with Dr. Ron Brookmeyer, *Biostatistics Department*, UCLA: Statistical methods for estimating HIV incidence from cross-sectional surveys of biomarker prevalence.
- 2012 – 2015     **Statistician**, *Surgery Department*, Stanford University School of Medicine: Served as shared statistical resource for department, focusing on health services research and effectiveness of new surgical imaging tools. Worked with large-scale state- and national-level AHRQ HCUP administrative data sets.
- 2010 – 2011     **Biostatistician/Bioinformatician**, *Aviir, Inc.*, Irvine, CA: Prediction of acute cardiovascular events using novel biomarkers; ELISA assay calibration.

## Peer-Reviewed Journal Articles

1.     Hitchner E, Zayed MA, Lee G, Morrison D, Lane B, Zhou W. **Intravascular ultrasound as a clinical adjunct for carotid plaque characterization**. *J Vasc Surg*. 2014;59(3):774–s80.

2. Wapnir I, Dua M, Kiernan A, Paro J, Morrison D, Kahn D, et al. **Intraoperative Imaging of Nipple Perfusion Patterns and Ischemic Complications in Nipple-Sparing Mastectomies.** *Ann Surg Oncol.* 2014;21(1):100–6.
3. Mell MW, Wang NE, Morrison D, Hernandez-Boussard T. **Interfacility transfer and mortality for patients with ruptured abdominal aortic aneurysm.** *J Vasc Surg.* 2014; 60(3), 553-557.
4. Moghavem N, Morrison D, Ratliff JK, Hernandez-Boussard T. **Cranial neurosurgical 30-day readmissions by clinical indication.** *J Neurosurg.* 2015;123(1):189–97.
5. Yanamadala S, Morrison D, Curtin C, McDonald K, Hernandez-Boussard T. **Electronic health records and quality of care: an observational study modeling impact on mortality, readmissions, and complications.** *Med (United States).* 2016;95(19):e3332.
6. DeClerck MP, Bailey Y, Craig D, Lin M, Auerbach LJ, Linney O, Morrison D, Patry W, Auerbach PS. **Efficacy of topical treatments for *Chrysaora chinensis* species: A human model in comparison with an in vitro model.** *Wilderness Environ Med.* 2016;27(1):25–38.
7. Pershing S, Morrison D, Hernandez-Boussard T. **Cataract surgery complications and revisit rates among three states.** *Am J Ophthalmol.* 2016;171:130–8.
8. Hitchner E, Morrison D, Liao P, Rosen A, Zhou W. **Genetic polymorphisms influence cognition in patients undergoing carotid interventions.** *Int J Angiol.* 2016;25(03):168–73.
9. Hernandez-Boussard T, Morrison D, Goldstein BA, Hsia RY. **Relationship of Affordable Care Act Implementation to Emergency Department Utilization Among Young Adults.** *Ann Emerg Med.* 2016;67(6):714–20.
10. Hernandez-Boussard TM, McDonald KM, Morrison D, Rhoads KF. **Risks of adverse events in colorectal patients: population-based study.** *J Surg Res.* 2016;202(2):328–34.
11. Hernandez-Boussard T, Morrison D, Tian L, Basu S. **Disparities in Healthcare Coverage and Utilization After Expanded Dependent Coverage.** *J Health Dispar Res Pract.* 2017;10(4):8.
12. Morrison D, Laeyendecker O, Konikoff J, Brookmeyer R. **Cross-Sectional HIV Incidence Estimation with Missing Biomarkers.** *Stat Commun Infect Dis.* 2018;10(1).  
<https://doi.org/10.1515/scid-2017-0003>
13. Laeyendecker O, Konikoff J, Morrison D, Brookmeyer R, Wang J, Celum C, et al. **Identification and validation of a multi-assay algorithm for cross-sectional HIV incidence estimation in populations with subtype C infection.** *J Int AIDS Soc.* 2018;21(2).  
<https://doi.org/10.1002/jia2.25082>
14. Morrison D, Laeyendecker O, Brookmeyer R. **Cross-sectional HIV incidence estimation in an evolving epidemic.** *Stat Med.* 2019;38(19):3614–27. <https://doi.org/10.1002/sim.8196>
15. Shetty V, Morrison D, Belin T, Hnat T, Kumar S. **Development and feasibility testing of a scalable system for passively monitoring oral health behaviors in the home setting.** *JMIR Mhealth Uhealth.* 2020;8(6): <https://doi.org/10.2196/17347>
16. Elmore JG, Wang PC, Kerr KF, Schrager DL, Morrison DE, Brookmeyer R, Pfeffer MA, Payne TH, Currier JS. **Excess Patient Visits for Cough and Pulmonary Disease at a Large US Health System in the Months Prior to the COVID-19 Pandemic: Time-Series Analysis.** *J Med Internet Res* 2020;22(9):
17. Essalat M, Morrison D, Kak S, Chang, E, Roig Penso I, Kulchar R, Madrid Padilla O, Shetty V. **A naturalistic study of brushing patterns using powered toothbrushes.** *PLoS One.* 2022 May 19;17(5):e0263638. <https://doi.org/10.1371/journal.pone.0263638>
18. Morrison D, Laeyendecker O, Brookmeyer R. **Regression with Interval-Censored Covariates: Application to Cross-Sectional Incidence Estimation.** *Biometrics.* 2022 Sep;78(3):908-21.  
<https://doi.org/10.1111/biom.13472>
19. Brookmeyer R, Morrison D. **Estimating Vaccine Effectiveness from Linking Population-Based Health Registries: Some Sources of Bias.** *American Journal of Epidemiology.* 2022

- Nov;191(11):1975-80. <https://doi.org/10.1093/aje/kwac145>
20. Grant-McAuley W, Laeyendecker O, Monaco D, Chen A, Hudelson SE, Klock E, Brookmeyer R, Morrison D, Piwowar-Manning E, Morrison CS, Hayes R. **Evaluation of multi-assay algorithms for cross-sectional HIV incidence estimation in settings with universal antiretroviral treatment.** BMC infectious diseases. 2022 Dec;22(1):1-1.
  21. Morrison DE, Nianogo R, Manuel V, Arah O, Anderson N, Kuo T, Inkelas M. **Modeling COVID-19 infection dynamics and mitigation strategies for in-person K-6 instruction.** Frontiers in Public Health. 2023 Feb 7;11:856940. <https://doi.org/10.3389/fpubh.2023.856940>
  22. Morrison DE, Ponzini MP, Santos ER, Biag HMB, Espinal G, Tassone F, Rivera SM, Hessel D, Schneider A, Bourgeois JA, Hagerman A, Kim K. **Progression of fragile X-associated tremor/ataxia syndrome revealed by subtype and stage inference.** Brain Communications. Published online 2025/12/10. <https://doi.org/10.1093/braincomms/fcaf483>

## Manuscripts Under Review

23. Narang K, Loomba V, Morrison DE, Vodonos J, Downey AE. **Associations Between Intimate Partner Violence and Purging Behaviors in Adolescents: Sex-Specific Insights From a Nationally Representative Dataset.** Journal of Interpersonal Violence, submitted 2024/12/02.
24. Seidman JC, Aiemyjoy K, Adnan M, Dehraj IF, Iqbal J, Iqbal K, Irfan S, Islam N, Kabir MS, Katuwal N, Maria N, Memon MA, Munira SJ, Naga SR, Pradhan S, Sarkar A, Shrestha R, Shrestha S, Al Sium SM, Vaidya K, Morrison DE, Carter AS, Saha S, Tamrakar D, Yousafzai MT, Garrett DO, Luby SP, Qamar FN, Saha S, Andrews JR, Charles RC. **Evaluating the accuracy of Salmonella Typhi Hemolysin E and lipopolysaccharide IgA to discriminate enteric fever from other febrile illnesses in South Asia.** medRxiv [Preprint]. 2025 Jun 22:2025.06.20.25329792. doi: [10.1101/2025.06.20.25329792](https://doi.org/10.1101/2025.06.20.25329792). PMID: 40585162; PMCID: PMC12204246. Emerging Infectious Diseases, submitted 2025/05/22.
25. Saiprom N, Nimnuan-ngam S, Phunpang R, Schildhauer SH, Morrison E, Tandhavanant S, Chandna A, Sengye S, West TE, Burtnick MN, Brett PJ. **Longitudinal Profiling of Antibody Responses to Burkholderia Pseudomallei: Implications for Diagnosis and Surveillance.** Lancet Microbe, submitted 2025/06/19. <http://dx.doi.org/10.2139/ssrn.5312903>
26. Lai, Kristina W., Orwa, Chris, Seidman, Jessica C., Garrett, Denise O., Saha, Samir K., Tamrakar, Dipesh, Qamar, Farah Naz, Charles, Richelle C., Andrews, Jason R., Teunis, Peter, Aiemyjoy, Kristen, Morrison, DE. **serocalculator, an R package for estimating seroincidence from cross-sectional serological data.** International Journal of Epidemiology, submitted 2025/06/30. Preprinted on medRxiv: <https://doi.org/10.1101/2025.06.04.25328941>

## Conference Presentations (\*indicates presenting authors)

1. Morrison D\*, Desai M, Curtin C, Hernandez-Boussard, T. **A Comparison of Cross-Sectional and Longitudinal Analyses for Adverse Events Data.** Joint Statistical Meetings 2014, Boston, MA.
2. Hernandez-Boussard T\*, Morrison D, Tian L, Basu S. **Increased Racial Disparities in Healthcare Utilization after Expanded Dependent Insurance Coverage.** 2016 Annual Research Meeting of AcademyHealth, Boston, MA.
3. Morrison D\*, Brookmeyer R, Laeyendecker O. **Biomarker-Based Cross-Sectional Incidence Estimation with Incomplete Data,** Joint Statistical Meetings 2017, Baltimore, MD.
4. Morrison D\*, Laeyendecker O, Brookmeyer R. **Transporting Cross-Sectional Incidence Estimation Algorithms Between Populations,** Joint Statistical Meetings 2019, Denver, CO.
5. O'Leary, M\*, Stockwell, R\*, Morrison, D. **Skills-based grading: a novel approach to teaching**

**formal semantics**, Annual Meeting of the Linguistic Society of America 2021, online.

6. Morrison D\*, Kuo T, Nianogo R, Manuel V, Inkelas M, Arah O. **Modeling COVID-19 Dynamics and Program Interventions for K-12 School Re-Opening**, AcademyHealth Annual Research Meeting 2021, online.
7. Morrison D\*, Brookmeyer R. **Regression with Interval-Censored Covariates: Application to Cross-Sectional Incidence Estimation**, Invited session, Eastern North American Region (ENAR) International Biometric Society 2023 Spring Meeting, Nashville, TN.
8. Morrison D\*, **Regression with Interval-Censored Covariates**, Bridging Statistical Strategies for Censored Covariates (invited workshop), Banff International Research Station, Alberta, Canada, 2024-02-01.
9. Aiemjoy K\*, Morrison, DE\*, 2024 **Serocalculator: An open-source R package for estimating seroincidence from cross-sectional serosurveys**. Institute for Disease Modeling Annual Symposium, October 1, 2024.
10. Morrison D\*, **Regression with Interval-Censored Covariates**. Invited session, 2025 WNAR/IMS Annual Meeting Whistler BC, Canada, June 17, 2025.
11. Schildhauer S, Lee KH, Morrison DE, Aiemjoy K. **Serodynamics: An open-source R-package to model post-infection antibody kinetics**. EPIDEMICS 10, San Diego, CA. 2025/12/01.

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### Conference Posters (\*indicates presenting authors)

1. Desai A\*, Patton C\*, Le M\*, Lin N\*, Morrison D, Shetty V. **A Smartphone-based ePlatform for Passive Monitoring of Oral Health Behaviors**. UCLA Undergraduate Research Poster Day, 2019, Los Angeles, CA.
2. Morrison D\*, Manuel V, Inkelas, M, Kuo, T, Arah, OA, Anderson, N, Nianogo, R. **Modeling COVID-19 infection dynamics and program interventions for K-12 school re-opening**, Translational Science 2021, online.
3. Tobin J\*, Greer Z\*, Morrison D. **eRecruitment Strategies for the COVID-19 Era**. UCLA School of Dentistry Research Day, 2022, Los Angeles, CA.
4. Ponzini M\*, Ren Y, Taylor S, Morrison DE, Wilson MD, Kim K. **Multi-modal data integration using PCA with multivariate longitudinal latent class analysis for patient subgroup stratification**. Joint Statistical Meetings 2023, Toronto, Canada.
5. Narang K, Loomba V, Morrison DE, Vodonos J, Downey AE. **Intimate Partner Violence and Purging in Adolescents: A Dose-Dependent Association With Sex Differences**. American Academy of Child & Adolescent Psychiatry (AACAP) 2024 annual meeting, Seattle, WA.  
<https://doi.org/10.1016/j.jaac.2024.08.029>
6. Narang K, Loomba V, Morrison DE, Vodonos J, Downey AE. **Intimate Partner Violence and Purging in Adolescents: A Dose-Dependent Association With Sex Differences**. American Academy of Pediatrics (AAP) 2024 conference, Orlando, FL.
7. Lee KH, Schildhauer SH, Munroe C, Ryan ET, Chowdhry F, Qadri F, Khan AI, Aiemjoy K, Morrison E, Charles R. **Modeling antibody kinetics following Shigella infection using a two-phase Bayesian hierarchical framework**. US-Korea Conference, Atlanta, GA. August 2025.
8. Lee KH, Schildhauer SH, Munroe C, Ryan ET, Chowdhry F, Qadri F, Khan AI, Aiemjoy K, Morrison E, Charles R. **Modeling antibody kinetics following Shigella infection using a two-phase Bayesian hierarchical framework**. American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, Toronto, Canada. November 2025.
9. Lee KH, Schildhauer SH, Munroe C, Ryan ET, Chowdhry F, Qadri F, Khan AI, Aiemjoy K, Morrison E, Charles R. **Modeling antibody kinetics following Shigella infection using a two-phase Bayesian hierarchical framework**. EPIDEMICS 10, San Diego, CA. December 2025.

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## Instructional Workshops

1. Instructor, **Serocalculator Workshop**, following the 13th International Conference on Typhoid & Other Invasive Salmonellosis in Kigale, Rwanda, 2023-12-08.  
<https://www.coalitionagainsttyphoid.org/serocalculator-workshop/>

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## Collaborative Workshops

1. Participant, **Bridging Statistical Strategies for Censored Covariates**, Banff International Research Station for Mathematical Innovation and Discovery (BIRS), 2024/01.

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## Software

1. Morrison D, Brookmeyer R. **rwicc** (“Regression With Interval-Censored Covariates”), v0.1.3 released 2022-03-09 on the Comprehensive R Archive Network: <https://CRAN.R-project.org/package=rwicc>
2. Teunis P, Lai K, Orwa C, Aiemjoy K, Morrison D (2025). **serocalculator: Estimating Infection Rates from Serological Data**. R package version 1.3.0, <https://CRAN.R-project.org/package=serocalculator> .
3. Teunis P, Schildhauer S, Aiemjoy K, Morrison D (2025). **serodynamics: Modeling Longitudinal Immune Seroresponses to Pathogens**. R package version 0.0.0.9038, <https://ucd-serg.github.io/serodynamics/>

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## Websites

1. *Regression Models for Epidemiology*, online lecture notes for UC Davis Epidemiology 204: Quantitative Epidemiology III: Statistical Models. Last modified 2025-07-17. <https://d-morrison.github.io/rme/>; source code <https://github.com/d-morrison/rme>
2. **Wastewater** statistical process control dashboard. First deployed 2023-01-21; latest feature update 2023-01-24. <https://ctsipopulationhealth.shinyapps.io/wastewater/>; source code: <https://github.com/UCLA-PHP/wastewater>
3. **FluView** statistical process control dashboard. First deployed 2022-09-30; latest feature update 2023-03-14. <https://ctsipopulationhealth.shinyapps.io/FluView/>; source code: <https://github.com/UCLA-PHP/FluView>
4. **AltaMed** statistical process control dashboard for COVID vaccination and case rates in Los Angeles County cities and communities. First deployed 2021-07-29; latest feature update 2022-03-10. <https://morrison.shinyapps.io/AltaMedCharts/>; source code: <https://github.com/d-morrison/Altamed>
5. **Los Angeles Unified School District COVID Testing** statistical process control dashboard. (password-protected)
6. **Los Angeles Unified School District Absence Clusters** statistical process control dashboard. (password-protected)
7. **SVECalc** model of student infection rates as a function of vaccination rates: <https://morrison.shinyapps.io/SVECalc/>
8. **RegionalCOVIDSchoolSimulation** model of student and family infection dynamics as a function of epidemiological conditions and school policies: <https://agent-based-models.shinyapps.io/RegionalCOVIDSchoolSimulation/>; source code: <https://github.com/UCLA-PHP/school.epi.abm>



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## Grants and Contracts

### *Grants Active:*

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|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-2027    | \$58,401, Investigator, Registry of Arthroscopy for Meniscal Pediatric Pathology (RAMPP), Brian Haus, MD (Principal Investigator). Pediatric Orthopedics Society of North America #A243065. |
| 2025-2028    | \$487,605, Investigator, <b>Validation of Enteric Fever Serological Tools</b> , Gates Foundation #26-1688                                                                                   |
| 2025-present | 20% FTE, Investigator. <b>Health Analytics Core - Strategic health system initiatives</b> ; MOU with UCDH Innovation Technology (UCDH IT)                                                   |

### *Grants Completed:*

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-2025 | \$50,000, Multi-Principal Investigator, Modeling Correlations in Antibody Response Dynamics to Enhance Diagnostic Accuracy and Improve Infectious Disease Surveillance, Kristen Aiemjoy (Principal Investigator), Convergence Research Award, UC Davis School of Medicine |
| 2023-2025 | \$422,496, Multi-Principal Investigator, Serocalculator: Estimating Incidence Rates from Serological Data, <b>EZRA MORRISON</b> , KRISTEN AIEMJOY (Multi-Principal Investigators), NIH National Institute of Allergy and Infectious Diseases (NIH/NIAID #1R21AI176416)    |
| 2022-2024 | \$3,000,205, Investigator, Impact of COVID-19 testing and mitigation on equitable return-to-school in the second largest US school district, MOIRA INKELAS (Principal Investigator), NIH/NICHD                                                                            |
| 2022-2024 | \$72,128, Collaborator, Investigating alternative methods for regression analysis for MIC data of bacterial bovine respiratory disease pathogens, HEATHER FRITZ (Principal Investigator), California Department of Food and Agriculture (CDFA #A21-0166).                 |
| 2017-2018 | \$36,000, Mentee, UCLA Graduate Research Mentorship: Measuring HIV Incidence Using Multi-Biomarker Assay Algorithms.                                                                                                                                                      |

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## Honors and Awards

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| 2026 | <a href="#"><u>Award for Excellence in Service to Graduate Students</u></a> , UC Davis Graduate Student Association                                |
| 2021 | <b>Carolbeth Korn Prize</b> for most outstanding graduating student, UCLA Fielding School of Public Health: \$10,000.                              |
| 2020 | <b>Celia and Joseph Blann Fellowship</b> for academic excellence and commitment to public health, UCLA Fielding School of Public Health: \$10,000. |

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## Teaching

### *At UC Davis:*

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 Spring | <b>Primary Instructor</b> , Epidemiology 204: <i>Quantitative Epidemiology III: Statistical Models</i> . <i>Student Evaluations</i> : “Overall Teaching Effectiveness:” 4.4/5; “Overall Educational Value of the Course:” 4.2/5. |
| 2024 Spring | <b>Primary Instructor</b> , Epidemiology 204: <i>Quantitative Epidemiology III: Statistical Models</i> . <i>Student Evaluations</i> : “Overall Teaching Effectiveness:” 4.8/5; “Overall Educational Value of the Course:” 4.8/5. |
| 2024 Winter | <b>Primary Instructor</b> , Epi 299: <i>Special Study for Epidemiology Graduate Students</i> .                                                                                                                                   |
| 2023 Fall   | <b>Primary Instructor</b> , Epi 299: <i>Special Study for Epidemiology Graduate Students</i> .                                                                                                                                   |
| 2023 Spring | <b>Primary Instructor</b> , Epidemiology 204: <i>Quantitative Epidemiology III: Statistical Models</i> . <i>Student Evaluations</i> : “Overall Teaching Effectiveness:” 4.5/5; “Overall Educational Value of the Course:” 4.4/5. |

2023 Spring    **Primary Instructor**, BST 299: *Special Study for Biostatistics Graduate Students*.  
 2023 Winter    **Primary Instructor**, BST 299: *Special Study for Biostatistics Graduate Students*.  
 2022 Fall        **Primary Instructor**, BST 299: *Special Study for Biostatistics Graduate Students*.

*At UCLA:*

2017 Winter    **Teaching Assistant**, Biostatistics 100B: *Introduction to Biostatistics*.  
*Student Evaluations:* “Overall Rating:” 8.5/9.

## Guest Lectures

1.    2023-10-02: **Modeling COVID-19 infection dynamics and mitigation strategies for in-person K-6 instruction**. EPI 290: Epidemiology Seminars, UC Davis.
2.    2023-03-02: **Introduction to Shiny Apps in R**. MPM 207: Applied Epidemiology Problem Solving, UC Davis. Handout: <https://ezra-morrison.shinyapps.io/Intro-to-Shiny>
3.    2022-11-04: **Agent-Based Models of COVID-19 System Dynamics in Schools**. EPI 203: System Science Methods for Population Health, UCLA.
4.    2022-04-26: **Regression modeling with interval-censored covariates: an EM algorithm approach**. BST 290: Graduate Group in Biostatistics Seminar, Department of Public Health Sciences, UC Davis.
5.    2022-01-19: **Introduction to Causal Inference**. Applied Statistics for Translational Researchers series, Clinical and Translational Science Center, UC Davis Health. Video: <https://ucdhs.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=43e9eb6f-3ed9-41ac-8ad9-ae22016572c8>; Slides: <https://health.ucdavis.edu/media-resources/ctsc/documents/pdfs/causal-inference-intro-2022.pdf>
6.    2023-11-09: **Causal inference challenges in an observational study of early childhood nightlight use and myopia**, California Epidemiologic Investigation Service (Cal-EIS) Fellowship professional development seminar series, Department of Public Health, UC Davis.
7.    2023-10-02: **Modeling COVID-19 infection dynamics and mitigation strategies for in-person K-6 instruction**, EPI 290: Seminars in Epidemiology, UC Davis.
8.    2024-02-12: **Introduction to Computational Programming with Julia**, UC Love Data Week; with Derek Devnich, Carl Stahmer, and Nick Ulle. Online, open-access course reader: [https://ucjug.github.io/workshop\\_julia\\_basics/](https://ucjug.github.io/workshop_julia_basics/)
9.    2024-11-08: **Agent-Based Models of COVID-19 System Dynamics in Schools**. EPI 203: System Science Methods for Population Health, UCLA.
10.    2025-01-23: **Causal inference challenges in an observational study of early childhood nightlight use and myopia**, California Epidemiologic Investigation Service (Cal-EIS) Fellowship professional development seminar series, Department of Public Health, UC Davis.
11.    2025-02-11: **Estimating incidence rates with cross-sectional biomarker data**. BST 290: Graduate Group in Biostatistics Seminar, Department of Public Health Sciences, UC Davis.
12.    2025-11-14: **Agent-Based Models of COVID-19 System Dynamics in Schools**. EPI 203: System Science Methods for Population Health, UCLA.
13.    2026-01-22: **Causal inference challenges in an observational study of early childhood nightlight use and myopia**, California Epidemiologic Investigation Service (Cal-EIS) Fellowship professional development seminar series, Department of Public Health, UC Davis.

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## Mentorship

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| 2024 – 2025    | Kush Narang, undergraduate, UC Davis. Project: Modeling associations between intimate partner violence and purging behaviors in adolescents. Presented at poster at American Academy of Child & Adolescent Psychiatry (AACAP) 2024 annual meeting, Seattle, WA. Submitted a manuscript to the Journal of Interpersonal Violence for peer review, 2024/12/02. Kush graduated from UC Davis in Spring 2025 with a Bachelor of Science degree in Computational Human Biology. In his final year at UC Davis, he won a <b>Civic Engagement Award</b> , a <b>Alys Hay Travel Award</b> , and a <b>CBS Distinguished Scholar Award</b> , and he received an <b>Honorable Mention for the 2025 Chancellor’s Award for Excellence in Undergraduate Research</b> . |
| 2024 – 2025    | Glorietta Kundetti, PhD student, Graduate Group in Epidemiology, UC Davis. Project: modeling antibody serodynamics. Project: Assessing Housing Stability and Viral Suppression Outcomes in the Housing Plus Program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2024 – present | Samuel Schildhauer, PhD candidate, Graduate Group in Epidemiology, UC Davis. Project: modeling scrub typhus antibody serodynamics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2023 – 2025    | Ramon Vara, PhD Student, Graduate Group in Epidemiology, UC Davis. Project: assessing effects of Don’t Ask, Don’t Tell on LGBTQ veterans’ health.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2023 – present | Kristina Lai, Doctoral candidate, Graduate Group in Epidemiology, UC Davis. Project: developing new statistical software for infectious disease incidence estimation from cross-sectional serosurveys.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2025 – present | Krista Vaidya, PhD candidate, Graduate Group in Epidemiology, UC Davis. Project: Characterizing the burden of pediatric scrub typhus in Kathmandu valley, Nepal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2022 – 2025    | Alec Michael, Doctoral candidate, Graduate Group in Epidemiology, UC Davis. Project: Modeling trends in antimicrobial resistance in poultry. Co-chair of dissertation committee. Graduated December 2025.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2022 – present | Kwan Ho Lee, Doctoral candidate, Graduate Group in Epidemiology, UC Davis. Project: Modeling antibody kinetics following Shigella infection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2022 – 2024    | Derek Pham, undergraduate, UCLA. Project: Data dashboards for COVID-19 outbreaks in K-12 school districts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2022           | Julia Tobin, dental student, UCLA. Project: analyzing recruitment strategies for dentistry research. Produced a poster presentation at the UCLA School Of Dentistry Research Day (Tobin et al, 2022).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2021           | Warren Coons, undergraduate, UCLA. Project: COVID-19 data dashboards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2018 - 2019    | Anika Desai, Chloe Patton, Maia Le, Nicole Lin; undergraduates, UCLA. Project: collecting usability and accuracy data for a mobile health app and system for recording oral health behaviors. Produced a poster presentation at the UCLA Undergraduate Research Poster Day (Desai et al, 2019).                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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## Doctoral Dissertation Committees

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|-------------------|----------------------------------------------------------------------|
| 2026/03 – present | Committee Member, Krista Vaidya, Graduate Group in Epidemiology      |
| 2026/03 – present | Committee Member, Samuel Schildhauer, Graduate Group in Epidemiology |
| 2025/01 - present | Committee Co-Chair, Kwan Ho Lee, Graduate Group in Epidemiology      |
| 2025/01 - present | Committee Co-Chair, Kristina Lai, Graduate Group in Epidemiology     |
| 2024/07 – present | Committee Member, Karen Click, Graduate Group in Epidemiology        |
| 2023/09 - present | Committee Member, Dingning Liu, Graduate Group in Biostatistics      |
| 2023/07 - present | Committee Member, Zhikuan Quan, Graduate Group in Biostatistics      |



2023/06 – 2025/12    Committee Co-Chair, Alec Michael, Graduate Group in Epidemiology (graduated)

### Doctoral Qualifying Exam Committees

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| 2025-11-17 | Committee Member, Zhengyu Lu, Graduate Group in Epidemiology              |
| 2025-11-12 | Committee Member, Maria Luisa Klobongona, Graduate Group in Epidemiology  |
| 2024-03-12 | Committee Member, Katy Shea, Graduate Group in Epidemiology               |
| 2023-09-26 | Committee Member, Dingning Liu, Graduate Group in Biostatistics           |
| 2023-06-03 | Committee Member, Zhikuan Quan, Graduate Group in Biostatistics           |
| 2023-04-21 | Committee Member, Peerawat Chintrakulchai, Graduate Group in Epidemiology |
| 2022-06-22 | Committee Member, Jingwei Xiong, Graduate Group in Biostatistics          |

### Service: Academic Committees

*At UC Davis:*

|                |                                                                            |                                                                                                          |
|----------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 2025           | GGE Epidemiology Learning Exchange                                         | Faculty advisor                                                                                          |
| 2024 - 2025    | Department of Public Health Sciences<br>Structures and Processes Workgroup | Junior Faculty Wishlist Subcommittee<br>Member                                                           |
| 2024           | Graduate Studies Fellowship Review<br>Committee                            | 17 fellowships reviewed                                                                                  |
| 2024 – present | GGE Admissions and Fellowships Committee                                   | 25 applications reviewed in 2026<br>30 applications reviewed in 2025<br>43 applications reviewed in 2024 |
| 2024 – present | GGE Exam Committee                                                         | 2 pre-qualifying exams co-written and graded                                                             |
| 2022 – present | GGE Educational Policy Committee                                           | IOR for PhD/MS-level core course Epi 204                                                                 |
| 2023           | GGPHS MPH Admissions Committee                                             | 40 applications reviewed                                                                                 |
| 2023 – present | GGPHS Teaching Committee                                                   | Member                                                                                                   |
| 2023           | GGB Exam Grading Committee                                                 | 2 exams graded                                                                                           |
| 2023           | Department of Public Health Sciences<br>Research Strategy Workgroup        | Workgroup member; managed workgroup<br>planning document                                                 |

### Chaired Sessions

**Strategies and Implications for Establishing Biostatistics Entities in Medical Centers.**

Invited Panel, Section on Statistical Consulting, Joint Statistical Meetings 2014, Boston, MA.

### Service: Ad-Hoc Journal Article Reviews

|             |                              |    |
|-------------|------------------------------|----|
| 2023        | Annals of Applied Statistics | 1x |
| 2022        | Epidemiology                 | 1x |
| 2022        | BMC Public Health            | 1x |
| 2021 - 2022 | Scientific Reports           | 2x |
| 2021        | Annals of Applied Statistics | 2x |

### Service: Ad-Hoc Workshop Reviews

|      |                                                                                          |    |
|------|------------------------------------------------------------------------------------------|----|
| 2022 | Banff International Research Station for<br>Mathematical Innovation and Discovery (BIRS) | 1x |
|------|------------------------------------------------------------------------------------------|----|

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## Professional Affiliations

- 2010 - present    **Member**, American Statistical Association (ASA)
- 2022 - present    **Member**, International Biometric Society, Western North American Region (WNAR)
- 2021 - 2022       **Member**, AcademyHealth
- 2022 – present   **Member**, UC Davis Julia User Group (D-JUG)